

Ekamveer Singh

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RESEARCH INTERESTS

Reliability and robustness of multi-model / multi-agent LLM systems; verification and escalation strategies for reducing hallucination and error in cascaded LLM pipelines; on-device and privacy-preserving inference.

EDUCATION

Punjab Engineering College

B.Tech, CSE-AI (CGPA: 9.01) + Minor, Quantum Technology (CGPA: 10.00)

Chandigarh, India

Expected May 2028

PROJECTS

LLM Cascade Eval | Ollama, Gemini API, Python, LLM-as-Judge

2026

- Designed a weak-to-strong multi-model cascade for code generation: problems escalate through progressively stronger reviewers, each shown the current draft and its exact failure reason, and only continue upward when the reviewing model does not confirm the draft is correct.
- Built an evaluation harness scoring code problems by direct test-suite execution and open-ended problems via a blinded, rubric-based third-party judge excluded from both systems under test, with a disclosed manual audit sample.
- Benchmarked against a single-shot call to Gemini 3.6 Flash; in an honest 3-problem pilot the cascade passed 3/3 versus 2/3, while reducing token consumption by ~61% (23K $\rightarrow$ 9K tokens) through adaptive escalation and verification. ([repo](#))

Gwen – Local Personal AI Assistant | MLX, MLX-Whisper, SpeechBrain, Ollama, SwiftUI Apr. 2026 – Present

- Architected a four-tier model routing system (on-device to cloud) with automatic fallback, and a query-routing layer built specifically to ground responses and reduce hallucination.
- Implemented speaker verification (ECAPA-TDNN) and acoustic echo cancellation (WebRTC AEC3) for reliable, full-duplex, barge-in-capable conversation.
- Built a fully local, privacy-first pipeline on Apple Silicon with multimodal I/O (image/PDF/DOCX), semantic memory, and persistent state; all code written and committed independently across 15 progressive stages. ([repo](#))

Automated Fungal Keratitis Screening | Python, FastAPI, OpenCV, Scikit-learn

July 2026

- Led system architecture for a 3-person team building a computer vision pipeline to screen fungal keratitis from confocal microscopy images, motivated by a clinical need raised by a PGI-MER physician to faculty.
- Built the LLM-based layer translating a stacked ensemble classifier's outputs and patient risk factors into clinician-readable diagnostic summaries; benchmarked across seven image-processing pipelines.

RESEARCH & PUBLICATIONS

Co-author, "Interface-Driven Electrochemical Performance of Hybrid MXenes for High-Performance Supercapacitors: Insights and Challenges" – *Current Opinion in Solid State & Materials Science*, under review. Independently authored the AI methodology section.

TECHNICAL SKILLS

Languages: Python, C++, Swift, SQL

ML / AI: PyTorch, TensorFlow, MLX, MLX-VLM, Scikit-learn, CatBoost, OpenCV

Systems: MLX Swift, SwiftUI, Apple Silicon inference, FastAPI, Ollama

Data: NumPy, Pandas, Matplotlib, SQL (MySQL, PostgreSQL, MongoDB, SQLite)

ACCOMPLISHMENTS

Finalist, Amazon ML Summer School 2026 | Published op-ed, *The Tribune* (2023)